CS166- Final Project

Minerva University
Prof. L. Tambasco

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1 Background

Historically, forests in California have been prone to wildfires, often caused by lightning striking brittle, dry bark. While to some extent a necessary process for the ecosystem, as some trees only reproduce after fires, damage to nearby civilization has been a longstanding issue. This paper aims to model wildfires in the Santa Monica Mountains, Los Angeles, California, USA, which have experienced notable fires in 2018 and 2024.

By modeling the spread of fire under various wind conditions, this paper aims to understand which parts of the map are especially prone to burning and considers the efficacy of possible interventions, specifically the placement of fire brigades and planned thinning.

2 Model

The model used in this report is inspired by Tambasco, 2025 with the following key extensions:

- **Wind:** The model takes into account the strength and direction of wind to calculate the probability of ignition of neighboring cells. Upwind neighbors are less likely to ignite, whereas downwind neighbors have a higher probability of catching fire. Wind has one of the following cardinal directions: N, NE, E, SE, S, SW, W, NW.
- **Tree density:** The probability of ignition also depends on tree density, with denser patches considered to have more 'fuel' and a higher chance of catching fire.
- **Data:** The model is based on canopy density data collected by Hansen et al., 2013 in 200.

2.1 Model Rules

The LA forest is modeled as a 200x200 cell grid with fixed (absorbing) boundaries and Moore neighborhoods. Each cell has an initial state of 'tree' (density > 0) or 'empty' (density $= 0$).

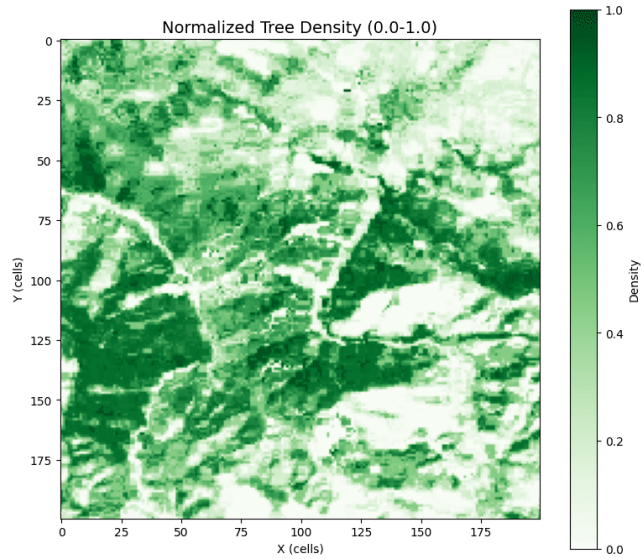

Figure 1: Density map of 6×6 km forest area. Darker colors indicate higher tree canopy density.

At the beginning of the simulation, unless a specific ignition start point is selected, a point is chosen at random to be ignited, changing the cell state from 'tree' to 'fire'.

Cells that are currently on 'fire' can ignite their neighbors. After one step of being on 'fire', the cell becomes 'char'red and can no longer spread the fire. Once the fire starts in a cell, on each step, it spreads stochastically to neighboring cells according to a probability based on

$$(\text{Base probability}) \times (\text{Density}) \times (\text{Wind factor}),$$

each found as:

- **Base probability:** The baseline 'combustibility' of cells under maximum density. This parameter can be considered as general fuel potential per tree, where high values could reflect drier wood and air in the winter months.
- **Density:** The density of the cell that the fire is trying to spread to. This is encoded by Hansen et al., 2013 and takes values between 0.0 and 1.0. Less dense cells thus have a lower probability of ignition.

- **Wind factor:** The multiplicative effect of wind, found by multiplying the alignment (the cosine of the wind's angle and the neighbor's angle from the source) and wind speed, adding 1, then clamping to 0.5 to avoid overly penalizing upwind cells.
  - Wind speed can take values from 0.0-2.0.
  - For example, if an east wind (east $\rightarrow$ west) of speed 2 is blowing, the wind factor for the NW (upper left) neighbor becomes 2 (speed) $\times \sqrt{2}$ (cosine of angle between wind and neighbor, in this case 45 degrees) ≈ 2.41. Since this is > 1, it increases the probability that the NW neighbor ignites, reflecting the directional bias introduced by wind. See more examples in Figure 2

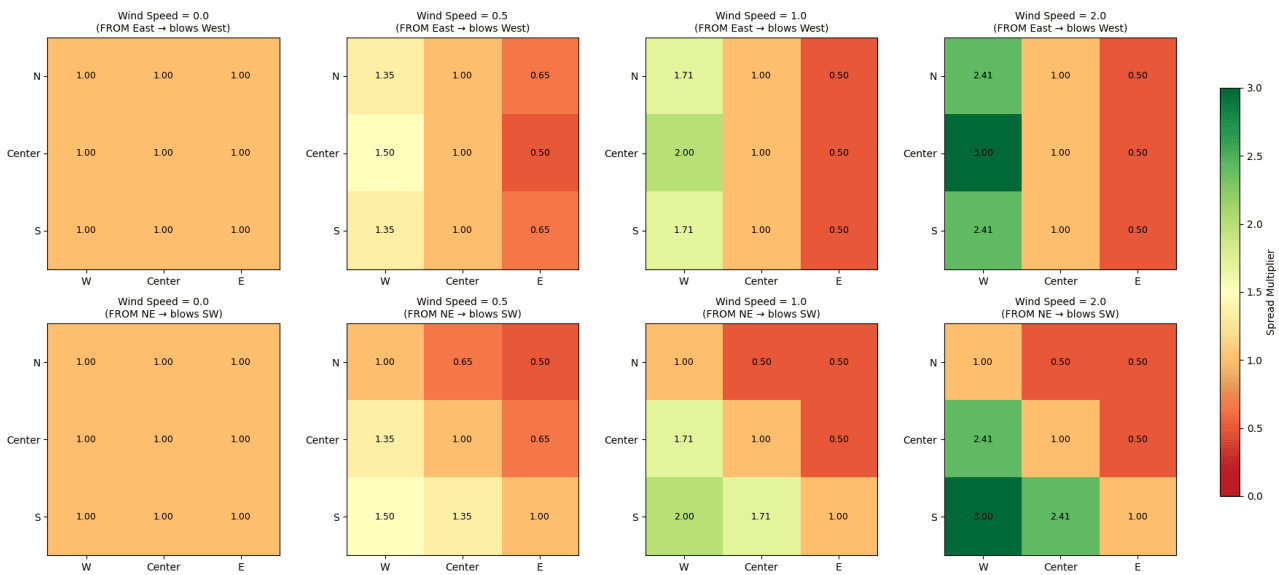

Figure 2: Wind factor for different wind strengths under East wind and NE wind.

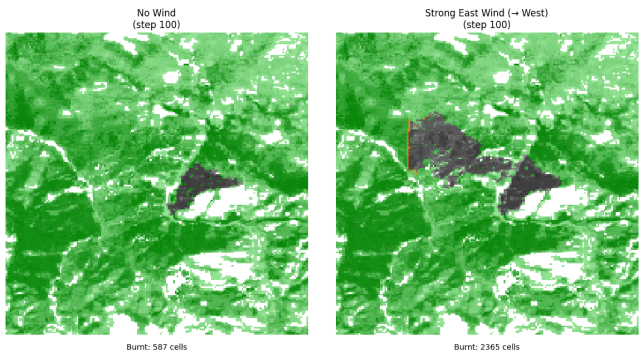

Figure 3: State of forest after 100 steps under no wind (left) and East wind (right). Darker greens represent higher tree density. Orange represents burning cells. Darker grays indicate higher-density cells that have burned. See accompanying zipped folder for GIF animation.

2.2 Model Outputs

After each run, the model returns the final configuration, a 2D numpy array that records the state of each cell.

By aggregating these across multiple simulations, we can find the following measures of risk:

- **Ignition risk per cell:** We define 'risk' as the likelihood that a cell has a burning neighbor at some point. This can be found as the frequency that each cell has a cell in its neighborhood that has an end state of 'fire' or 'char'. It can be used as a measure of risk for the cell to be impacted by the fire, either by being directly ignited or by being adjacent to a burning cell at some point. This was then normalized to values between 0.0 and 1.0 by the total number of runs.
- **Burnt area:** This is found as the number of cells that are burnt or burning at the end of a single run. It represents the size of the affected area.
- **Fire spread:** This is the Euclidean distance from the initial ignition point to the furthest burnt or burning cell at the end of a single run.
- **Percolation risk:** Since the starting point is random, we define percolation as the fire spreading beyond a 1 km radius from the initial ignition point, rather than the fire spanning the entire grid. Thus, if the fire spreads beyond 1 km, we consider it to have percolated.

3 Results

To understand fire behavior, we ran the simulation 200 times each for nine scenarios: winds from the north, northeast, east, southeast, south, southwest,

west, and northwest, and no wind to serve as a baseline. For scenarios with wind, we set wind strength to 1.0 to generate comparable results.

The following table summarizes the results:

Table 1: Monte Carlo Results (400 Steps)

Condition	Mean Burn (cells)	95% CI	Percolation (%)	Mean Dist (km)
No Wind	8875.0	[7947.9, 9802.0]	73.0	3.31
0° (N)	6779.1	[6005.6, 7552.6]	67.0	2.97
45° (NE)	7398.1	[6677.1, 8119.2]	76.0	3.23
90° (E)	6753.7	[5926.4, 7580.9]	71.0	2.98
135° (SE)	7342.3	[6443.1, 8241.5]	75.0	3.03
180° (S)	7444.1	[6456.2, 8432.0]	68.5	3.00
225° (SW)	7004.4	[5985.1, 8023.7]	73.0	2.87
270° (W)	7127.6	[6181.3, 8073.9]	75.5	3.17
315° (NW)	7340.4	[6409.6, 8271.3]	70.0	3.17

3.1 Individual Cell Risk

We visualized risk as a heatmap, where darker colors indicate higher risk. From Figure 4, it can be seen that under no wind, risk roughly corresponds to tree density. From Figure 4, it can be observed that wind creates a bias in the direction it is blowing, increasing risk in areas downwind and decreasing it for areas upwind. It can also be observed that lower-density areas (such as the center-right in the figures) or densely forested areas insulated from the center by a strip of sparsely wooded land (such as the bottom and top right) tend to have a lower risk, regardless of wind conditions.

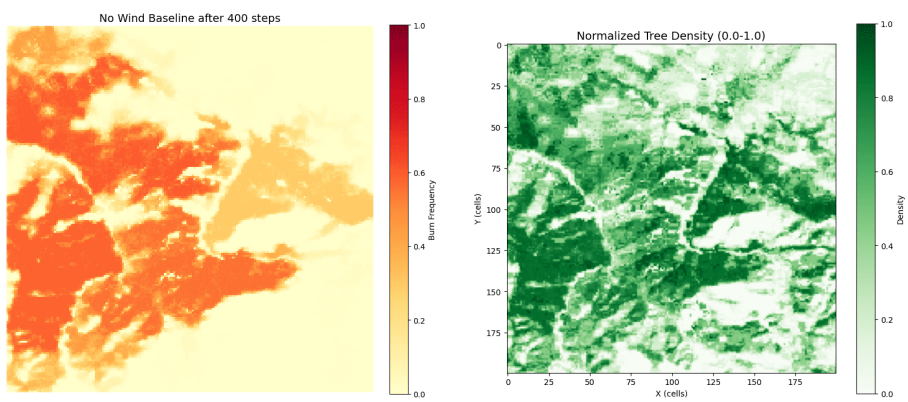

Figure 4: Left: Risk heat map of the LA forest area under no wind. Darker colors indicate higher risk. Right: Density map.

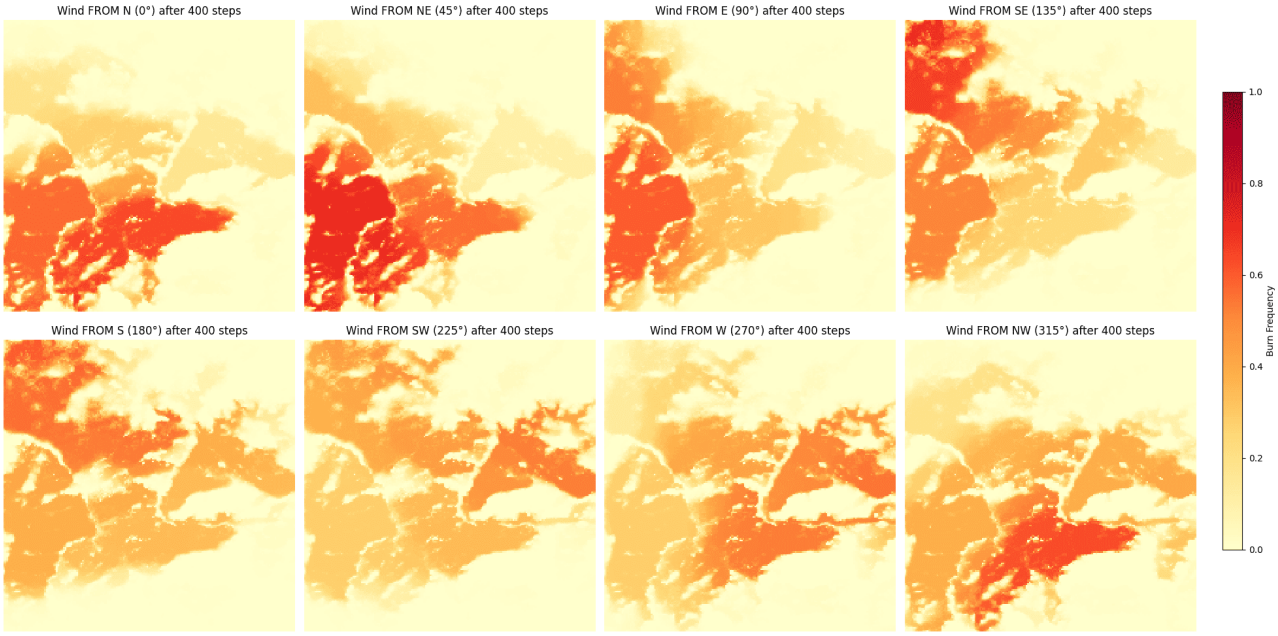

Figure 5: Risk heatmap of LA forest area under different wind conditions. Darker colors indicate higher risk.

We also analyzed the distribution of risk across cells. Figure 6 shows a bimodal distribution, where the majority of cells are at lower risk, and some are at very high risk. Figure 7 suggests that wind has different effects, depending on direction: winds from the N, NE, E, and SE increase the number of low-risk cells, but also increase the maximum risk observed on the whole grid. This matches the heatmap, as we can see more extreme values (indicated by darker shades of red) in these wind conditions. Conversely, winds from the S, SW, W, and NW seem to introduce another peak with somewhat high risk, but decrease the number of low-risk cells.

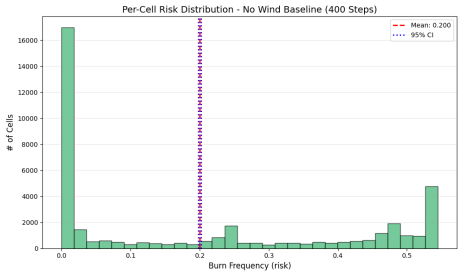

Figure 6: Histogram of risk distribution under no wind.

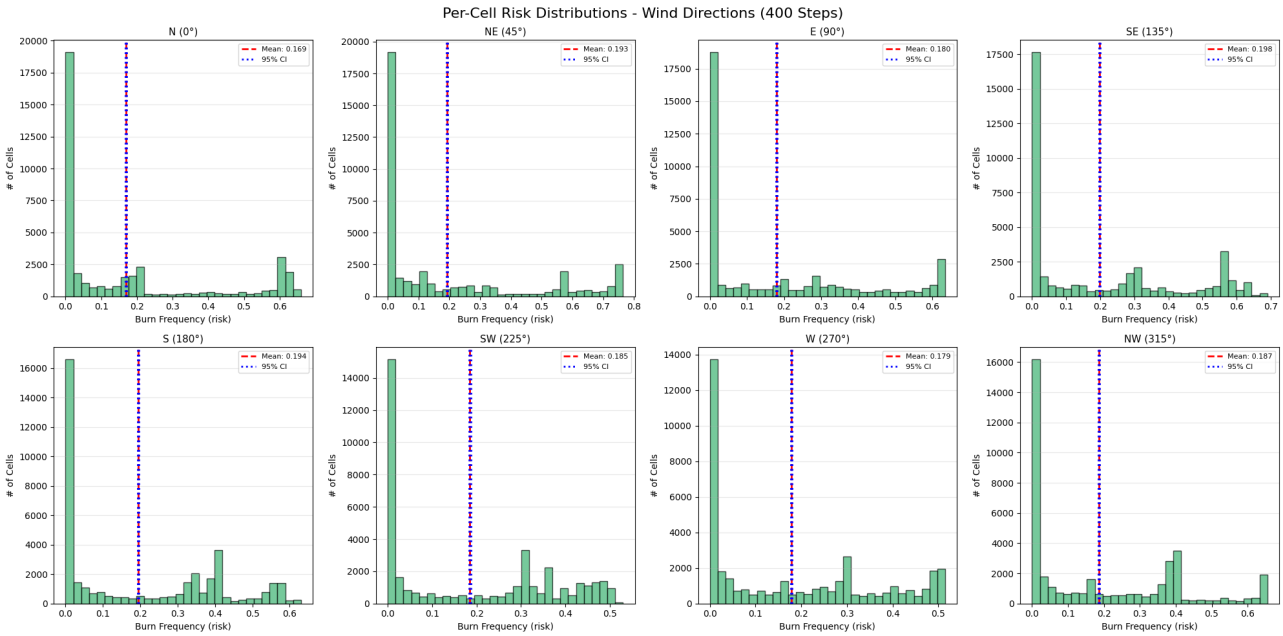

Figure 7: Histogram of risk distribution under different wind conditions.

3.2 Burnt Area

We can also consider the distribution of the number of burnt cells over 200 runs. Figure 8 suggests that, under no wind, the fire dies out without spreading around 25% of the time. However, when the fire does spread, it spreads drastically, as indicated by the clustering at the right side of the histogram. This suggests the existence of a 'tipping point', where if the fire reaches a certain densely forested region, it is much more likely to spread to a larger part of the map.

On the other hand, Figure 9 suggests that with wind, the fire is slightly more likely to spread to an intermediate number of cells. This can be attributed to wind causing directional bias, which can overcome fires dying out in localities with low density, allowing them to spread to more regions. However, the frequency of fires contained in a small area remains similar, indicating that only the distribution (and not the frequency) of fires burning larger areas was affected.

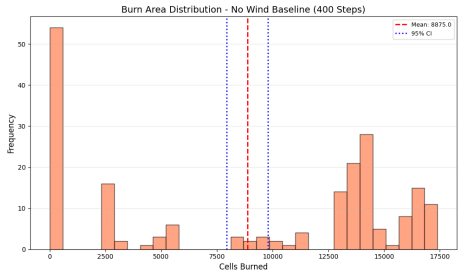

Figure 8: Histogram of burnt area under no wind.

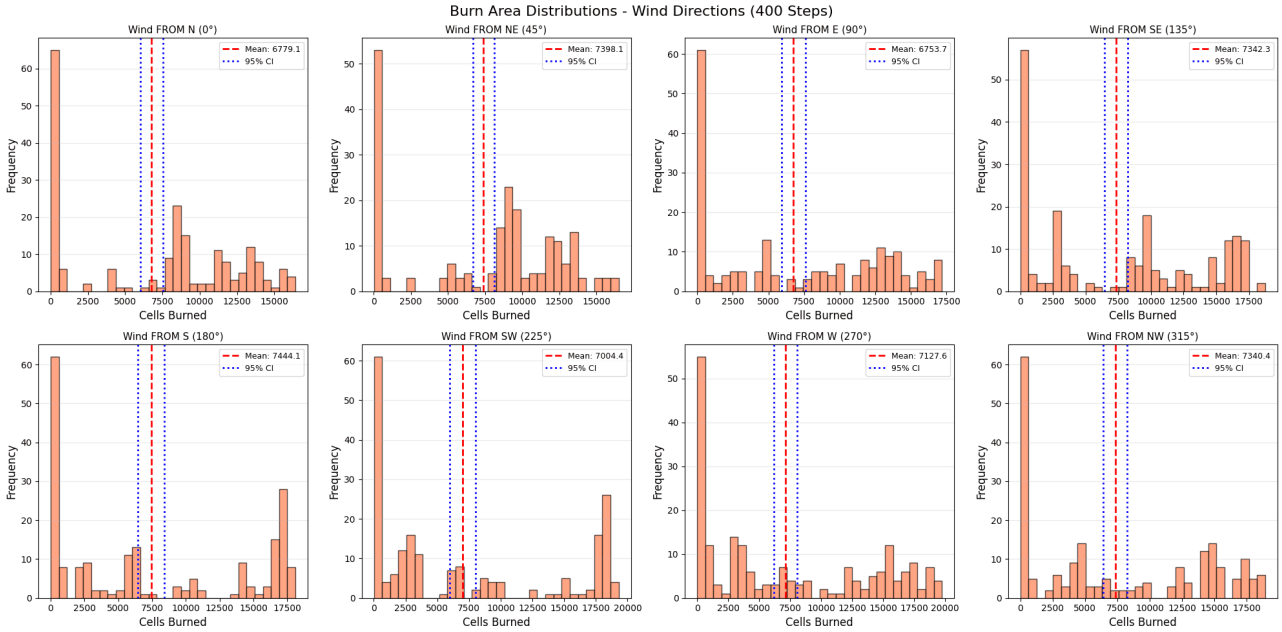

Figure 9: Histogram of burnt area distribution under different wind conditions.

3.3 Fire Spread

Similarly to the burnt area, we can also consider the distribution of fire spread over runs. Figure 10 appears bimodal, suggesting that fires under no wind sometimes do not spread, and when they do, they tend to reach wide areas, which aligns with the assessment of burnt area above. From Figure 11, it can be inferred that the introduction of wind makes spread more likely, as the peak at the right side of the histogram becomes taller. Together with the analysis of burnt areas above, this suggests that wind tends to elongate the spread of fire

rather than increasing its overall spread.

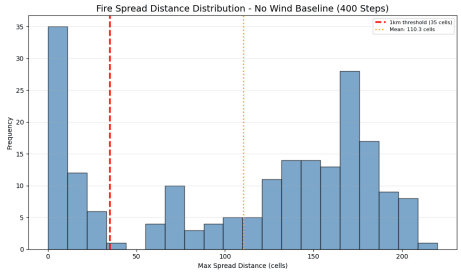

Figure 10: Histogram of fire spread under no wind.

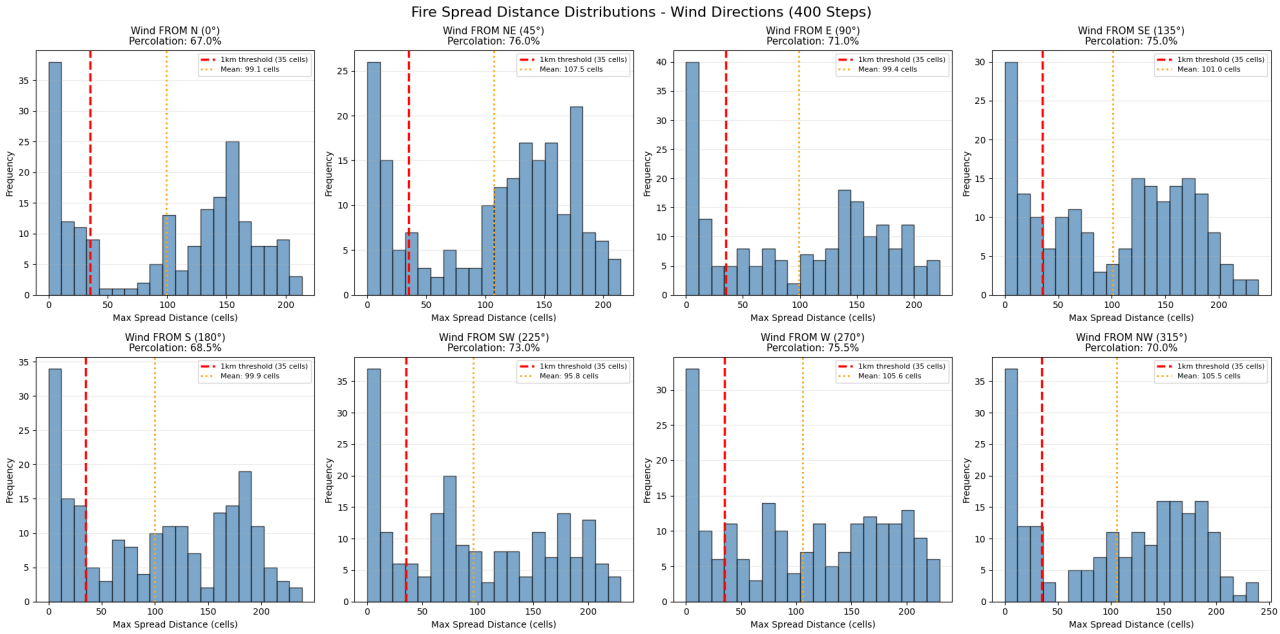

Figure 11: Histogram of fire spread distribution under different wind conditions.

3.4 Recommendations

Based on the analysis above, we can infer that while wind plays a significant role in determining the spread of fire, density remains a substantial determinant of risk. Thus, we suggest two interventions to prevent fires from spreading to wide areas and causing damage:

- **Planned thinning and installment of fire breaks:** As seen in Figures 4 and 5, density is a significant predictor of risk. Thus, to control fire outbreaks, we propose that the most dense parts of the forest be thinned, or that fire breaks be installed in the form of less flammable trees or planned clearings to prevent fires from percolating across the dense areas and spreading to the entire map.

- **Targeted deployment of patrol and fire-fighting resources based on weather forecasts:** As seen in Figure 5, downwind regions face significantly higher risks. Thus, we recommend that fire forces prioritize patrolling these areas, especially during dry seasons or after thunderstorms, to detect fires early and prevent their spread.

4 Analyzing uncertainty

The Monte Carlo simulations with 200 trials per condition yield confidence intervals of varying width across the three key metrics. For per-cell risk (burn frequency), all conditions exhibit narrow uncertainty, with relative CI widths ranging from 0.4% to 1.1% of the mean. This is well below the 10% threshold for low uncertainty (see Figure 12). This precision is obtained by averaging across all 40,000 grid cells, which provides substantial statistical power even with 200 trials.

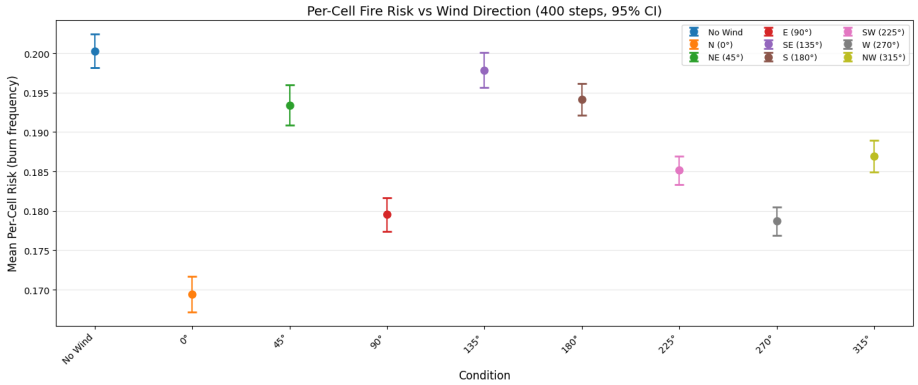

Figure 12: Error plot of individual cell risk. Error bars represent 95% confidence intervals.

In contrast, burn area measurements show much higher uncertainty, with relative CI widths ranging from 18.2% to 28.7% of the mean across conditions. Most conditions fall into the "WIDE - High uncertainty" category (>20% relative width), with the no-wind baseline exhibiting a CI width of 1877.9 cells (23.4% of mean) (see Figure 13). Wind conditions demonstrate similar or

higher uncertainty, with southwest and west winds showing the widest intervals at 28.7% and 28.0% respectively. Spread distance metrics exhibit moderate uncertainty, with relative CI widths between 16.4% and 20.5%, though several conditions still exceed the 20% threshold (see Figure 14).

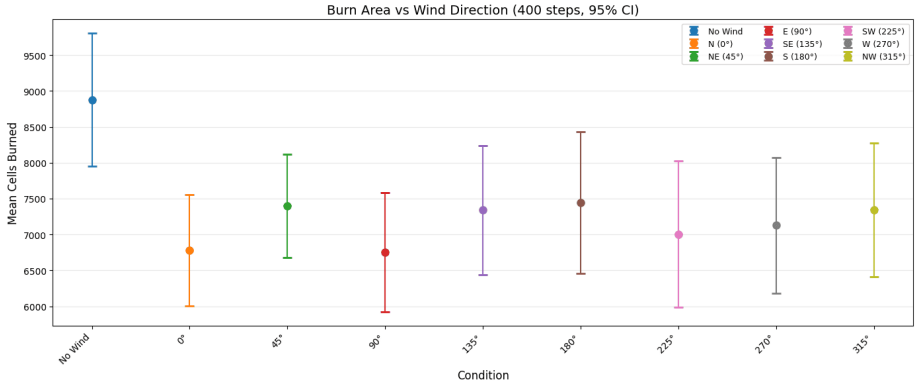

Figure 13: Error plot of burnt area. Error bars represent 95% confidence intervals.

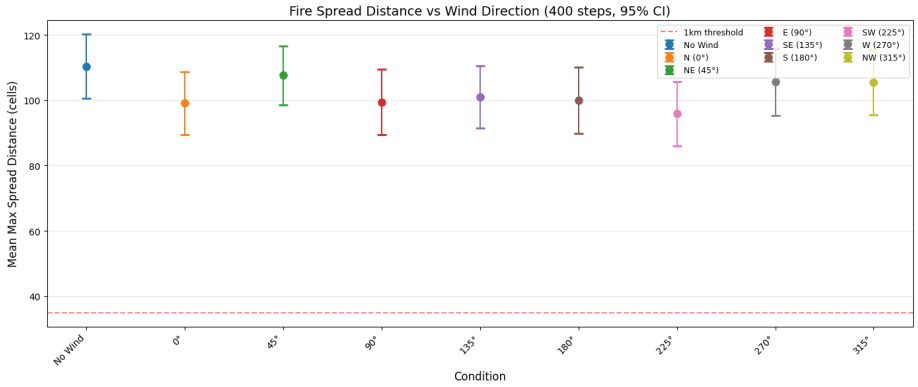

Figure 14: Error plot of fire spread. Error bars represent 95% confidence intervals.

To achieve narrower confidence intervals for the higher-uncertainty metrics, we will require substantially more trials. Since CI width scales with $1/\sqrt{n}$, halving the current burn area and spread distance interval widths would require approximately 800 total trials (600 additional runs beyond the current 200), while achieving quarter-width intervals, which will bringing these metrics into the "NARROW" category (<10% relative width), would demand roughly 3200

trials (3000 additional runs). Given that the current experiment with 200 runs per condition requires several minutes, scaling up to 3200 trials will take several hours. The per-cell risk estimates, already achieving narrow confidence intervals, would likely not see significant practical benefits from additional trials.

AI Statement:

I used ChatGPT (conversation link) for help exporting the tree density data, formatting, and advice on computational approaches. The initial idea to use matrix dilution in my risk mapping came from ChatGPT, and I consulted the SciPy documentation to confirm that it behaves as intended. I also used GitHub Copilot's auto-complete feature for coding support in my Python implementation.

References

Hansen, M. C., Potapov, P. V., Moore, R., Hancher, M., Turubanova, S. A., Tyukavina, A., Thau, D., Stehman, S. V., Goetz, S. J., Loveland, T. R., Kommareddy, A., Egorov, A., Chini, L., Justice, C. O., & Townshend, J. R. G. (2013). High-resolution global maps of 21st-century forest cover change [Data available online]. *Science*, *342*(6160), 850–853. http://earthenginepartners.appspot.com/science-2013-global-forest

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